

An End-to-End AI Agent for Kaggle Competitions

Benchmarked against two frozen reference agents on 23 Kaggle competitions

Wei-Hao Huang

Abstract

We present the **Sinica Kaggle Agent**, an end-to-end agent that runs a machine-learning competition from raw data to submitted prediction with no human in the loop. A large language model (LLM) reasons and decides at every stage while AutoML libraries are invoked through the shell, and a tree search over complete candidate solutions serves as the optimization loop. The agent was run in isolated sandboxes for all 23 competitions of the study (transcript-audited on the 20 ordinary ones) and benchmarked against two frozen, methodologically distinct reference agents, **AIDE** (LLM tree search) and the **NVIDIA Kaggle Agent** (public-kernel reproduction). The 23 competitions comprise 20 ordinary ones, whose verdicts come from real submissions scored on the live leaderboard, and 3 special ones spanning language, physiological time series, and medical imaging, graded offline with MLE-bench. **Among the 23 competitions, Sinica obtains the best performance, with 13 wins and 10 losses**, taking the best private score on 10 of the 20 ordinary competitions and the three-way best on all 3 special ones.

1 Introduction

A Kaggle competition scores every submission on a hidden test set and ranks thousands of teams by that number. Solving one has traditionally required expert labor and repeated trial and error, but because every attempt is scored, the process can be posed as an optimization loop and handed to a machine. LLM-driven coding agents have recently begun to automate this loop. **AIDE** performs LLM-guided tree search over candidate solutions [1]; the system of Aygün et al. couples an LLM rewriter with tree search and externally injected research ideas, outperforming strong baselines across 16 Kaggle Playground competitions [2]; and industrial agents skip search altogether by reproducing the strongest public solution to a given competition [3].

This report is about one such agent, **the Sinica Kaggle Agent**, a hybrid design in which an LLM reasons at every stage of the competition pipeline, AutoML libraries are invoked through the shell, a cross-competition experience library supplies evidence-backed priors, and a tree search over candidate solutions serves as the optimization loop (Figures 1 and 2). The agent also documents its own runs: each one writes a per-competition specification spanning data, models, training, inference, and evaluation.

To evaluate the Sinica agent, we benchmark it against two frozen reference agents, **AIDE** (LLM tree search with no cross-competition memory) and the **NVIDIA Kaggle Agent**, which reproduces the highest-voted public kernel. Sinica’s lanes ran in isolated sandboxes across all 23 competitions; on the 20 ordinary ones each lane’s transcript was audited (Section 3) and the final submission was scored on the real leaderboard via late submission, while the 3 special competitions are graded offline with MLE-bench [4]. Sinica takes the best score on 13 of the 23 competitions, the strongest standing of the three agents (Section 4).

2 Methods

2.1 The Agent

The method is implemented as a Claude Code Skill (`kaggle-agent`) around a division of labor. Execution follows the six stages of Figure 1 (data ingestion, EDA, feature engineering, modeling, evaluation, submission), with the tree search looping between modeling and evaluation. The LLM (Claude Code) handles problem understanding, validation-strategy design, feature engineering, result interpretation, and the choice of next step, while LightGBM / XGBoost / CatBoost / Optuna, invoked via the shell, handle model selection, hyperparameter optimization, and stacking/ensembling.

The injection layer, the purple elements of Figures 1 and 2, supplies external knowledge at the head of the pipeline through two components:

- **Stage 0.5, the problem dossier:** before EDA, the agent answers “what kind of problem is this?” by naming task family, train–test window relation, split policy, and external-data need in `dossier.json`, grounded in a structured library of evidence-backed task-level priors (`knowledge/knowledge_base.json`), rendered per competition by `knowledge/task_priors_for.py`. Reading competition-specific discussions or kernels is forbidden; this keeps the method distinct from reproduction-based agents such as the NVIDIA Kaggle Agent. The dossier remains a prior until EDA verifies its hypotheses and records a verdict.
- **External-data module:** external data come only from a whitelist of public sources, and a join rule guarantees that no row sees information from after its own time period. Every dossier proposal is tracked to an outcome, realized or logged as unrealized, so a failed injection cannot pass for a successful one. The layer’s measured effect is reported in the Appendix.

Two record-keeping mechanisms support the pipeline (Figure 1):

- **Experiment records:** every experiment’s parameters, features, split scheme, and scores are logged in structured form (`experiments.json`, `experiments_tree_v3.json`), and each lane records a structured exploratory-data-analysis (EDA) pass (distributions, drift, correlation) before modeling, which the skill’s reporting layer renders, with the dossier and experiment records, into a per-competition ML specification report (see the Appendix).
- **Cross-competition experience library** (`knowledge/experience.md`): every insight must carry an evidence field (`competition`, `exp #N`, `score A` $\rightarrow$ `score B`); entries without a score delta are rejected. The agent queries the library before EDA and before modeling, and writes validated new insights back after experiments. This is the main design difference between Sinica and the two reference methods.

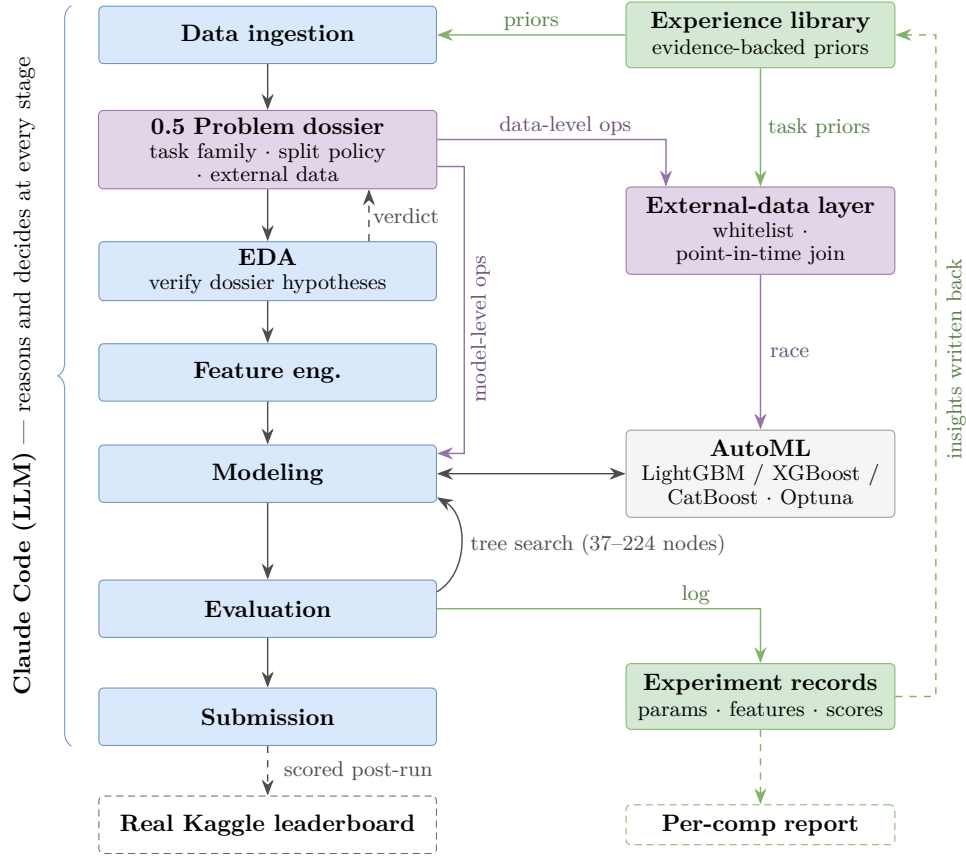


Figure 1: Architecture of the Sinica Kaggle Agent, with color marking roles. **Blue, the LLM-driven pipeline:** Claude Code reasons and decides at every stage. **Purple, the injection layer:** the dossier’s proposals enter as external-data arms or search lineages. **Green, the evidence loop:** the experience library and the experiment records. **Gray, AutoML tools** invoked via the shell. Dashed arrows mark feedback, advisory, or post-run steps.

Modeling and evaluation run in one of two optimization modes:

- **Linear iteration:** propose a hypothesis, run the experiment, record the result (used for the first pass and for small competitions).
- **Tree search:** once linear iteration has produced a single baseline model and at least one blend, the agent switches to tree search. Every ordinary-competition run in this report uses the tree-search harness of Figure 2 (`tree_search/harness_v3.py`).

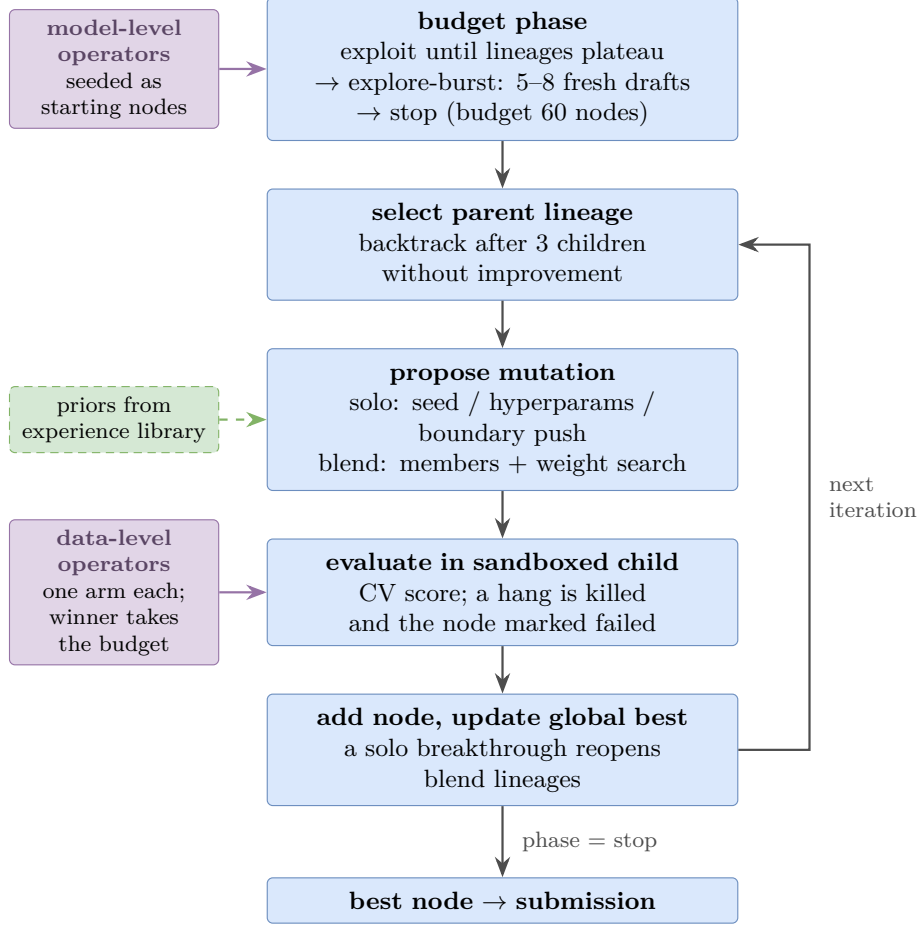


Figure 2: The tree-search loop used in all Sinica runs (default budget 60 evaluated nodes per competition; per-lane counts are in the experiment logs). Each node is a complete, executable solution scored by local cross-validation; blends over cached out-of-fold (OOF) predictions are first-class nodes, and the best node becomes the submission. Purple: the injection layer’s operator entry points; green: priors from the experience library; dashed arrows: advisory input and backtracking.

2.2 The Two Reference Agents

AIDE (AI-Driven Exploration). AIDE performs LLM-driven tree search: at each step, it writes a complete solution, executes it, reads the error or score, and mutates accordingly. It carries no cross-competition memory.

NVIDIA Kaggle Agent. The agent researches the competition context, selects the highest-voted solution-type kernel, ports it faithfully, and retrains on local data, reproducing the strongest public kernel.

Both reference methods are frozen because improving a reference agent mid-study would make earlier and later data incomparable and would quietly turn the benchmark into an optimization of the reference methods.

3 Benchmark Design

Evaluation set. The study uses 23 competitions in two groups. The 20 *ordinary* competitions are tabular Kaggle competitions with live late submission, comprising 7 classification tasks (binary and ordinal) and 13 prediction/regression tasks including time series and multi-target, with training sets from 1,157 to 630,000 rows (Table 1). The 3 *special* competitions sit outside the tabular family. They are us-patent

(NLP text pairing), ventilator-pressure-prediction (physiological time series), and SIIM-ISIC melanoma (dermoscopic imaging), prepared and graded offline by MLE-bench [4], which replays finished Kaggle competitions with their original data and metrics.

Table 1: Competition descriptions.

#	Competition	Task type	Train rows $\times$ cols
1	s3e1	Regression	$37,137 \times 10$
2	s3e3	Classification (binary)	$1,677 \times 35$
3	s3e5	Classification (ordinal)	$2,056 \times 13$
4	s3e7	Classification (binary)	$42,100 \times 19$
5	s3e9	Regression	$5,407 \times 10$
6	s3e11	Regression	$360,336 \times 17$
7	s3e14	Regression	$15,289 \times 18$
8	s3e16	Regression	$74,051 \times 10$
9	s3e19	Regression (time series)	$136,950 \times 6$
10	s4e11	Classification (binary)	$140,700 \times 20$
11	s5e10	Regression	$517,754 \times 14$
12	s6e1	Regression	$630,000 \times 13$
13	s6e2	Classification (binary)	$630,000 \times 15$
14	afsis-soil-properties	Regression (multi-target)	$1,157 \times 3,600$
15	cat-in-the-dat	Classification (binary)	$300,000 \times 25$
16	conway-s-reverse-game-of-life	Structured multi-output	$50,000 \times 802$
17	tps-aug-2022	Classification (binary)	$26,570 \times 26$
18	tps-jan-2022	Regression (time series)	$26,298 \times 6$
19	s5e1	Regression (time series)	$230,130 \times 6$
20	tps-sep-2022	Regression (time series)	$70,128 \times 6$
21	us-patent-phrase-matching	Regression (text similarity)	$32,825 \times 5$
22	ventilator-pressure-prediction	Regression (time series, seq-to-seq)	$5,432,400 \times 8$
23	siim-isic-melanoma	Classification (binary, imaging)	$28,984 \times 8$

Rows 1–20: the ordinary competitions (7 classification, 13 prediction/regression). Rows 21–23: the 3 special competitions; sizes are the MLE-bench re-split training sets.

Execution conditions. The three agents ran serially on one machine, each method as shipped. AIDE ran its default 20 steps, NVIDIA reproduced public kernels per its method, and Sinica ran the full six-stage pipeline including tree search (default budget 60 evaluated nodes per competition, extendable through the logged resume path). Four isolation conditions (data, workspace, instruction, and resource) are explicitly enforced (enforcement details in the Appendix).

Sandboxes, audit, and scoring. Each of Sinica’s 20 ordinary-competition lanes ran in an isolated root with the Kaggle token stripped, so the agent could neither consult the leaderboard nor submit. After a lane finished, an audit read every tool call in its transcript before the score was accepted. An accepted lane produces a final `submission.csv`, which we submit from outside the sandbox to the finished competition for the official public and private scores.

4 Results

4.1 The 20 Ordinary Competitions

Table 2 gives the complete benchmark, the public and private leaderboard scores of all three agents on the 20 competitions. **Sinica takes the best private score on 10 of the 20** (NVIDIA 7, AIDE 3).

Table 2: The complete 20-competition benchmark.

#	Competition	Metric	Sinica LB	NVIDIA LB	AIDE LB
1	s3e1	RMSE ↓	0.55275 /0.55715 (77.1)	0.55550/0.55332 (93.2)	0.55552/0.55899 (68.0)
2	s3e3	ROC-AUC ↑	0.87157/0.89449 (30.6)	0.89537 /0.93526 (66.8)	0.86863/0.88484 (26.4)
3	s3e5	QWK ↑	0.57223/0.57041 (72.0)	0.56974/0.57515 (75.7)	0.58138 /0.59362 (84.4)
4	s3e7	ROC-AUC ↑	0.90441/0.91127 (74.3)	0.90458 /0.91230 (78.1)	0.90109/0.90934 (67.1)
5	s3e9	RMSE ↓	12.33192/11.84750 (51.2)	12.22913 /11.83399 (59.1)	12.30667/11.86657 (44.5)
6	s3e11	RMSLE ↓	0.29368 /0.29309 (76.6)	0.29624/0.29559 (66.6)	0.29445/0.29389 (73.3)
7	s3e14	MAE ↓	333.18611/343.43434 (66.3)	330.71616 /338.36524 (89.2)	332.31556/342.44918 (71.2)
8	s3e16	MAE ↓	1.33822 /1.33738 (95.1)	1.34001/1.33566 (96.4)	1.34224/1.33920 (91.4)
9	s3e19	SMAPE ↓	49.40878/49.11872 (37.6)	48.26558 /47.52145 (41.4)	48.34131/48.39715 (39.9)
10	s4e11	Accuracy ↑	0.94107 /0.94195 (64.0)	0.94076/0.94136 (52.8)	0.94056/0.94184 (62.1)
11	s5e10	RMSE ↓	0.05576 /0.05554 (81.2)	0.05588/0.05562 (59.4)	0.05579/0.05555 (78.9)
12	s6e1	Error ↓	8.71516 /8.68844 (76.8)	8.73239/8.70480 (69.5)	8.72191/8.69449 (74.8)
13	s6e2	ROC-AUC ↑	0.95505/0.95354 (68.6)	0.95508/0.95348 (63.0)	0.95516 /0.95364 (76.5)
14	afsis	MCRMSE ↓	0.48917 /0.41902 (72.7)	0.69855/0.61276 (18.8)	0.49861/0.45636 (45.6)
15	cat-in-the-dat	ROC-AUC ↑	0.80243 /0.80800 (72.6)	0.77084/0.77641 (32.6)	0.80217/0.80790 (70.3)
16	conway	MAE ↓	0.10414 /0.10313 (100.0)	0.11189/0.11094 (96.5)	0.11065/0.10960 (97.9)
17	tps-aug-2022	ROC-AUC ↑	0.58930/0.58534 (63.5)	0.58730/0.57920 (43.0)	0.59106 /0.58479 (60.5)
18	tps-jan-2022	SMAPE ↓	6.29895/4.58859 (75.4)	4.63651 /4.43958 (76.4)	10.66638/10.11598 (28.9)
19	s5e1	MAPE ↓	0.08401 /0.09076 (74.3)	0.14594/0.05726 (84.4)	0.17917/0.12573 (61.6)
20	tps-sep-2022	SMAPE ↓	28.49605/28.45070 (13.6)	5.48351 /4.77458 (81.1)	9.72095/5.78998 (72.1)
Median PR (range)			72.7 (13.6–100.0)	68.2 (18.8–96.5)	69.2 (26.4–97.9)

Cells: private/public (PR) LB scores; **bold** = best *private* (direction-aware); PR = percentile rank of the public score against 2026-08-24 public-leaderboard snapshots.

4.2 The 3 Special Competitions

The three agents also ran the three special competitions. Because one node is a full deep-learning fine-tune rather than a gradient-boosting refit that takes seconds, a tree search over dozens of nodes would overrun the lane’s compute budget. Therefore, on each of the three competitions, Sinica only ran the linear-iteration mode of Section 2.1. It iterated to one validated model, then built and submitted a blend of the models it had trained. MLE-bench then grades each submission offline on its held-out private split. Sinica takes the three-way best on all 3 special competitions (Table 3). These lanes ran without tree search or external-idea injection, so winning across three unrelated task families points to pipeline discipline (validation designed before modeling, a structured EDA pass, every experiment logged with its evidence) as an advantage of its own.

Table 3: The 3 special competitions, graded by MLE-bench.

Competition	Metric	Sinica	NVIDIA	AIDE
us-patent	Pearson r ↑	0.86658 † (silver)	0.86160† (bronze)	0.65821
ventilator	MAE ↓	0.16525	0.40093	0.34591
SIIM-ISIC	AUC ↑	0.93529 †	0.89822	0.88579

Bold = best of the three; † = above the frozen leaderboard’s median.

4.3 Final Standing Across 23 Competitions

Counting the best score on each of the 23 competitions, **Sinica performs best, winning 13 and losing 10**. The other 10 wins, 7 by NVIDIA and 3 by AIDE, all come on ordinary competitions (Table 2).

Both of Sinica’s heaviest losses occur where a mature public solution exists. On tps-sep-2022, the public baseline is so strong that both reference agents beat Sinica by a wide margin (SMAPE 5.48 and 9.72 against 28.50); on tps-jan-2022, NVIDIA reproduced the well-known GDP-ratio solution and won 4.64 to 6.30.

5 Discussion

5.1 Comparing the Three Agents

The three agents differ in where knowledge comes from and in how the next step is decided rather than in tuning skill (Table 4).

Table 4: Design comparison of the three methods.

Aspect	Sinica	AIDE	NVIDIA
Knowledge source	Cross-competition experience library (accumulating)	None (from zero each time)	Public kernels fetched fresh each time
Search strategy	Six-stage pipeline + tree search (60-node default budget)	Tree-shaped trial and error, 20 steps	No search; faithful reproduction
Basis for next step	Experience-library priors + OOF diagnostics	Previous step’s error or score	Highest-voted kernel
Validation rigor	Split chosen by task type; diagnostics tagged and excluded	No concept of split legality	Inherits the kernel’s split
External data	Injection layer (whitelisted public sources)	Never considered	Inherited from the kernel
Run documentation	Writes its own five-section specification report per competition	None	None

The reference agents’ wins track their knowledge source. Sinica’s edge is validation rigor and accumulated knowledge (on s3e19 it tried the same shuffled KFold split that AIDE submitted from, but tagged it as a diagnostic and excluded it from submission), while its measured bottleneck was judgment at problem identification, the gap the injection layer was built to close.

5.2 Lessons from the Study

Running one agent through 23 competitions and reading every transcript also documents where the approach falls short. Each lesson below traces to a concrete incident or measurement from this study.

- **The agent is strong at coding but weak at judging.** The agent works through whatever problem it is given without asking whether the change is worth making, and the cost of a poor design choice surfaces only after results return (our first injection design, which injected knowledge downstream at the blend stage, is our own instance).
- **Local validation cannot stand in for the leaderboard.** Inside a lane the leaderboard is unreachable, so iteration optimizes cross-validation (CV) alone, and the two can diverge sharply: s3e19’s pooled OOF SMAPE of 6.71 sat against a realized 49.41. No local CV number is quoted as evidence anywhere in this report.
- **Isolation must be structural, not behavioral.** We stripped credentials, separated the lane roots, and, after the incident that taught us this lesson, mounted the experience library read-only. The one cross-lane

contamination in the study came from a shared file that was still writable, not from an agent breaking a rule it was given (incident log in the Appendix, Table 5).

- **Settle the architecture before implementation.** In our agent, external data originally entered at the final blending stage. In practice, weight search had already optimized that stage to its limit, so adding external data there changed almost nothing (the measured gain was negligible). We later moved the injection to the head of the pipeline, where it informs what kind of problem the agent faces and which external resources to bring in, and the scores improved (all six paired comparisons won; arm scores in the Appendix). But the move forced a redesign of the whole architecture, and every competition already run had to be rerun.
- **The agent’s prose needs human editing.** Its reports over-weight technical detail and reinvent terms draft after draft; every specification report attached to this study needed human editing.

5.3 Limitations

- **The sample is small and the standing is sensitive to its composition.** Reproduction-based and search-based methods win on disjoint kinds of competitions, so adding or removing a few competitions of one kind moves the win count. Reproduced kernels were also tuned by their authors against the realized public leaderboard, so the comparison is least symmetric exactly where the reproduction method is strongest.
- **The special-competition lanes ran linear iteration only, not the full tree-search method** (Section 4.2), and offline grading yields one score per submission, with no public/private pair to corroborate it.

5.4 Open Problems

Two problems remain open.

- **Choosing the form of an injection.** When the layer proposes external data, it must also choose how to use them, as a joined feature or as a target rescaling. No signal we tested makes that choice reliably.
- **The controlled evidence is thin.** Few competitions need external data at all, so the evidence rests on three competitions, and only two of them (tps-sep-2022 and s5e1) played no part in building the task-prior library the dossier consults. Finding more competitions that need external data is worth more than re-running ones where the layer stays silent.

6 Conclusion

All 23 competitions ran in isolated sandboxes; the 20 ordinary ones were transcript-audited and scored on the real leaderboard, and the 3 special ones were graded offline by MLE-bench. Sinica performs best among the 23, with 13 wins and 10 losses: it took the best private score on 10 of the 20 ordinary competitions and the three-way best on all 3 special ones. Its losses concentrate where a mature public solution exists and reproduction beats search.

Appendix: Experimental Procedure and Availability

Fairness protocol. Because the three agents share one machine, four isolation conditions are explicitly enforced. Two are enforced by tooling, one by directory structure, and one, instruction isolation, by a written protocol with an incident log (Table 5). The incident log records one cross-lane contamination: a lane read entries another lane had written back into the shared experience library, which the harness had left writable; the lane was killed and rerun clean after the library was reverted, and the library is now mounted read-only.

Table 5: The four isolation conditions and what enforces each.

Condition	Rule	Enforcement
Data	Each side reads only its own root, containing files validated against the official Kaggle manifest	<code>benchmark_infra/build_isolated_roots.py</code>
Workspace	The three output directories are fully separated; each side is scored independently	Directory structure
Instruction	Prompts may carry environment facts and metric thresholds only, never modeling recipes, another side’s scores, or a workaround it discovered	Manual protocol + incident log
Resource	Only one lane executes at any moment	<code>benchmark_infra/lane_lock.sh</code> mutex

Per-competition specification reports. A specification report accompanies this paper for each of the 23 Sinica runs, written by the agent at the end of the run it describes; neither reference agent documents its own runs (our collector re-documented AIDE’s runs after the fact). All follow a five-section ML specification framework (Data / Models & Architecture / Training / Inference / Evaluation & Benchmarking) and are archived in `docs/ml_specs/`. Every number in them is extracted from structured run records by collector scripts, never generated by an LLM, and unavailable fields are marked “not recorded.” The reports are CV-only; their conclusions must not be mixed with leaderboard verdicts.

External-data arm scores. The layer proposes external data only when a competition needs it; most do not. On the three competitions where it did (s3e19, tps-sep-2022, s5e1), we ran the pipeline with and without each proposed operator (`join_feature` and `ratio_target`) and submitted every run for a real leaderboard score, six paired comparisons in all. All six external-data arms beat their no-external baselines on the private leaderboard (Table 6).

Table 6: Per-arm private leaderboard scores on the three competitions where the layer proposed external data.

Arm	s3e19	tps-sep-2022	s5e1
no external data	54.506	25.476	0.15751
<code>join_feature</code>	48.497	23.390	0.15626
<code>ratio_target</code>	52.073	24.091	0.12417

SMAPE for s3e19 and tps-sep-2022, MAPE for s5e1; lower is better. **Bold** = best per competition. The s3e19 and tps-sep-2022 arms are matched configurations (identical pipeline, no search, no blending); the s5e1 arms ran under the full searched pipeline.

Execution environment. All lanes ran on ARM64 Linux (Ubuntu 24.04) with an NVIDIA GB10 GPU, under a uv-managed Python 3.13 environment frozen in `uv.lock`, with fixed seeds and LightGBM set to `deterministic=true`, `force_row_wise=true`, and a fixed `num_threads`; without these settings, the same configuration does not reproduce across processes.

Availability. All code and benchmark artifacts are archived in github.com/930731haohao-afk/kaggle-agent (private; access granted on request); every number in this report traces to those structured records.

References

- [1] Jiang, Z., Schmidt, D., Srikanth, D., Xu, D., Kaplan, I., Jacenko, D., and Wu, Y. AIDE: AI-Driven Exploration in the Space of Code. arXiv:2502.13138 (2025).
- [2] Aygün, E., et al. An AI system to help scientists write expert-level empirical software. *Nature* **654**, 909 (2026). <https://doi.org/10.1038/s41586-026-10658-6>
- [3] NVIDIA. Kaggle Agent: kernel-reproduction agent skill (software), 2026. <https://github.com/NVIDIA/nvidia-kaggle>
- [4] Chan, J. S., et al. MLE-bench: Evaluating Machine Learning Agents on Machine Learning Engineering. In *Proceedings of the 13th International Conference on Learning Representations*, 2025. <https://openreview.net/forum?id=6s5uXNWGIh>